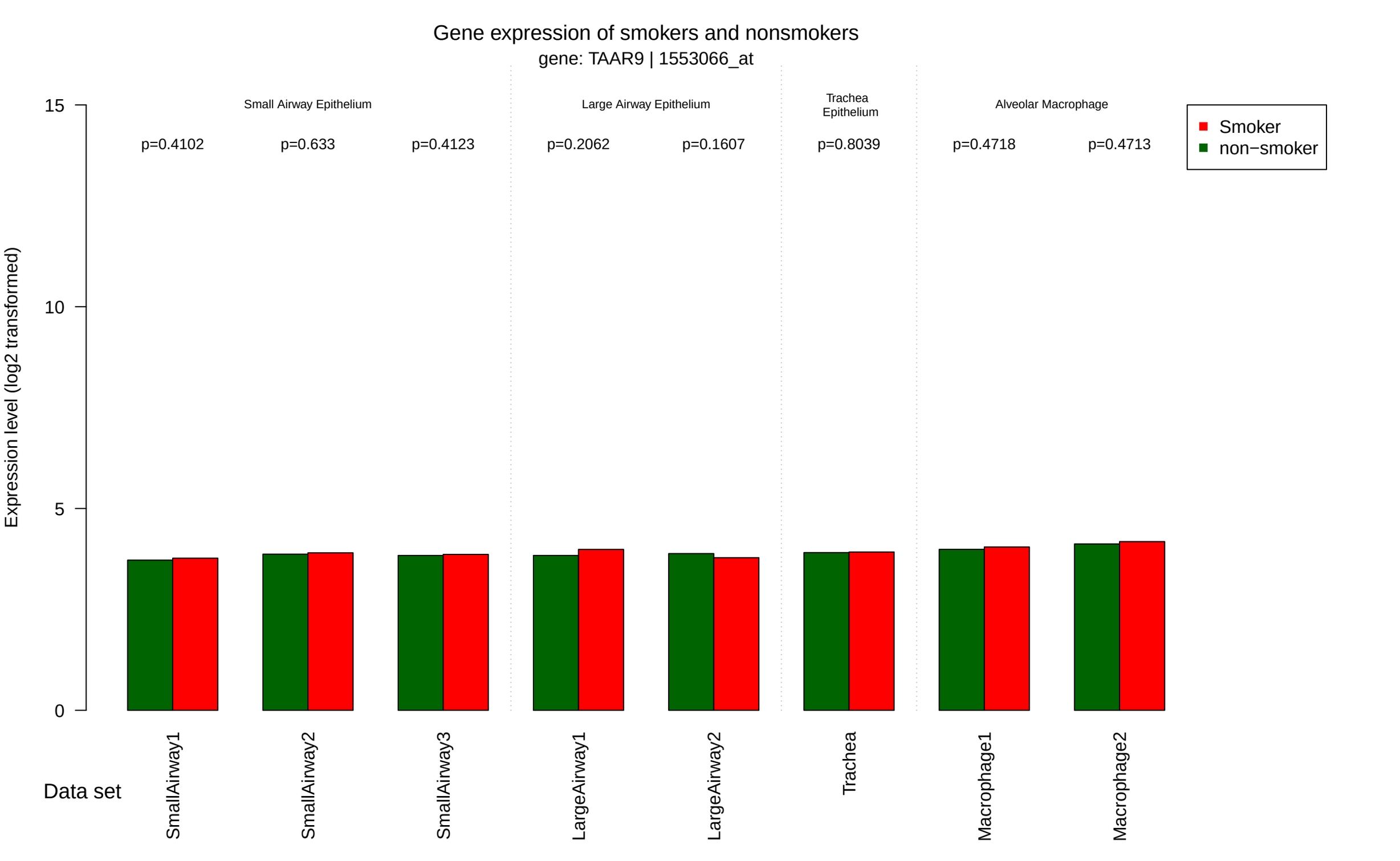
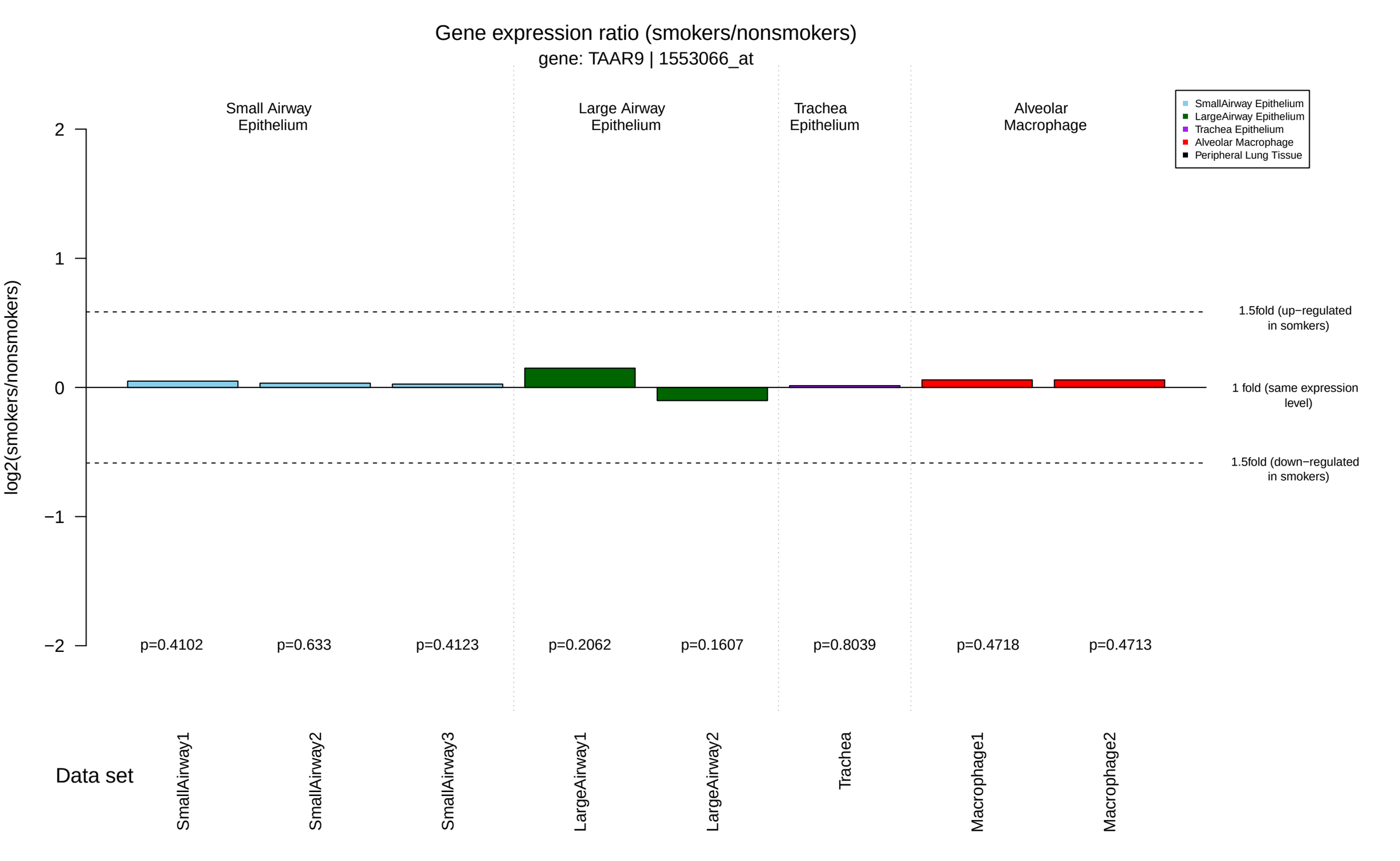
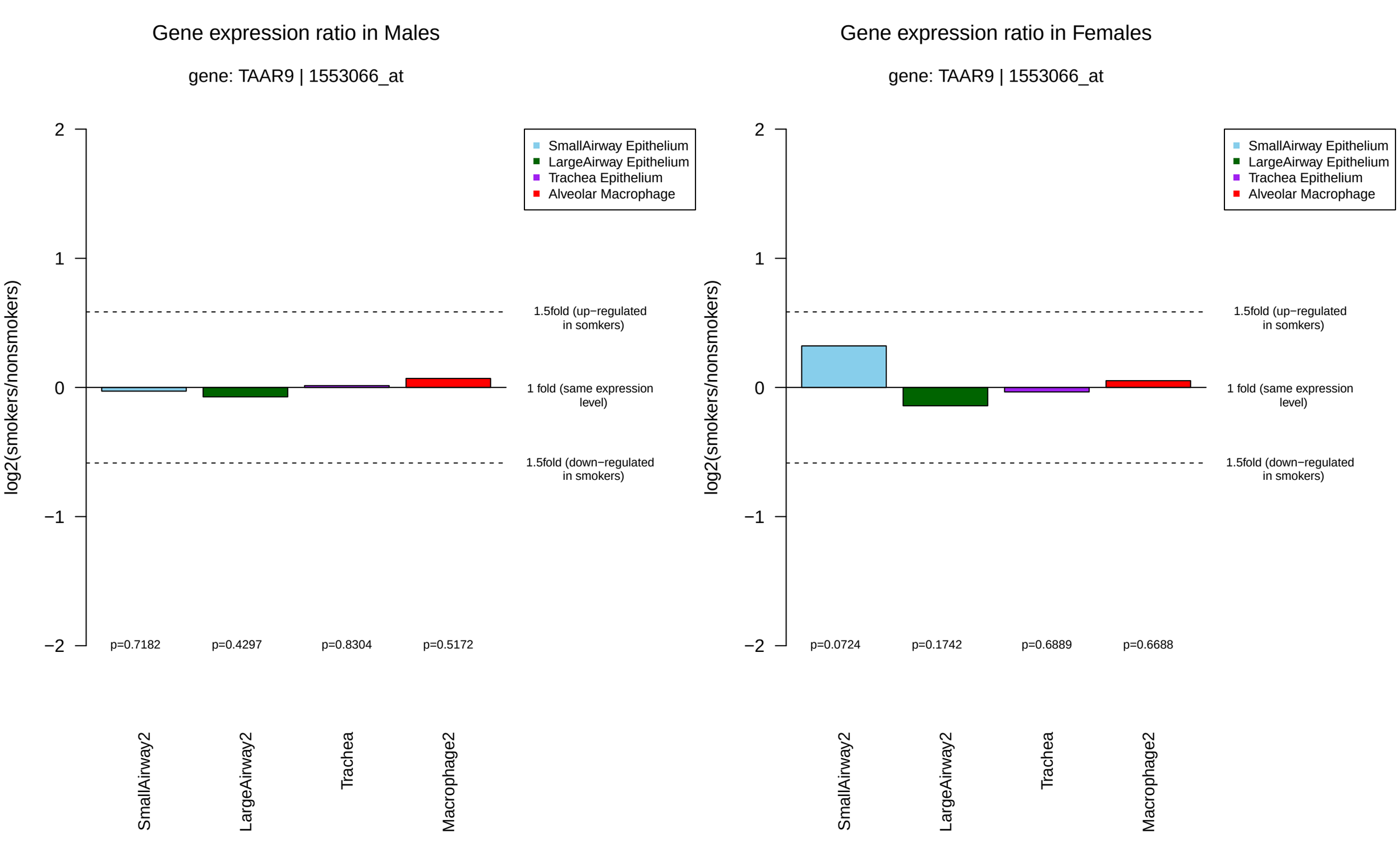
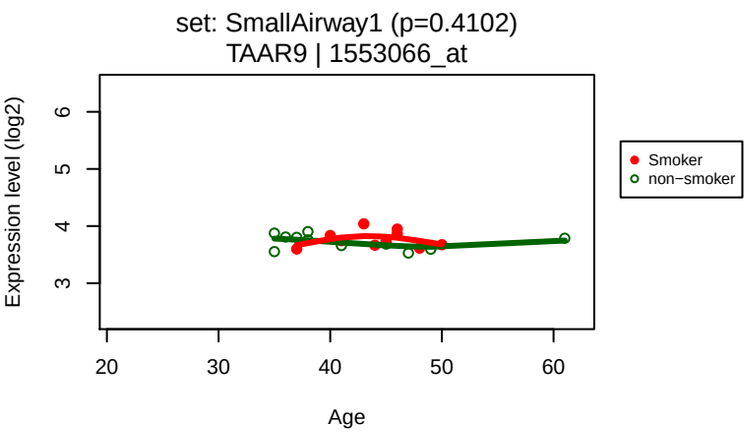
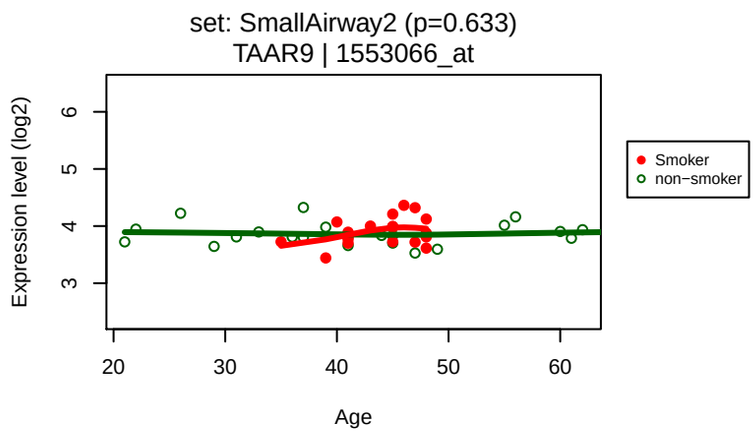
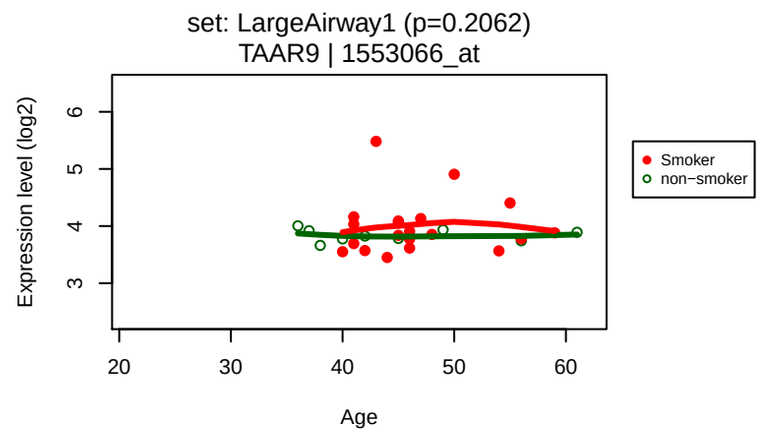

Gene expression of smokers and nonsmokers
gene: TAAR9 | 1553066_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4102
p=0.633
p=0.4123
p=0.2062
p=0.1607
p=0.8039
p=0.4718
p=0.4713
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

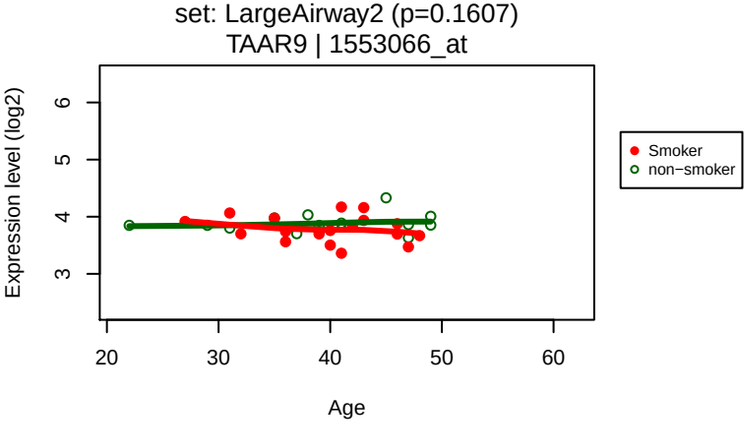

Gene expression ratio (smokers/nonsmokers)
gene: TAAR9 | 1553066_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4102
p=0.633
p=0.4123
p=0.2062
p=0.1607
p=0.8039
p=0.4718
p=0.4713
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

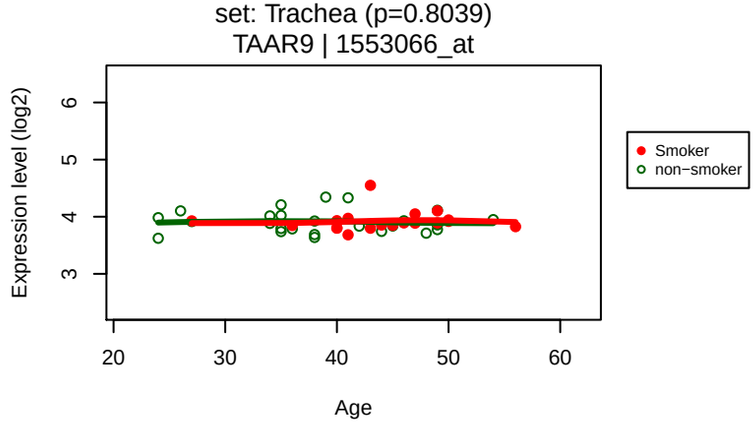

Gene expression ratio in Males
gene: TAAR9 | 1553066_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7182
p=0.4297
p=0.8304
p=0.5172
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAAR9 | 1553066_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0724
p=0.1742
p=0.6889
p=0.6688
SmallAirway2
LargeAirway2
Trachea
Macrophage2

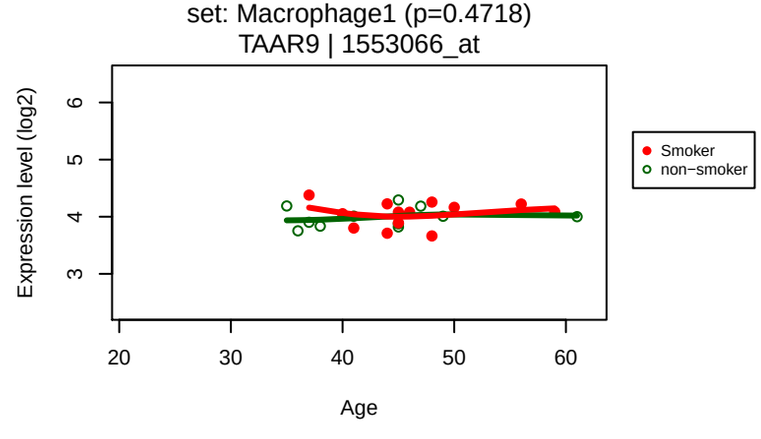

set: SmallAirway1 (p=0.4102)
TAAR9 | 1553066_at
Expression level (log2)
Age
Smoker
non−smoker

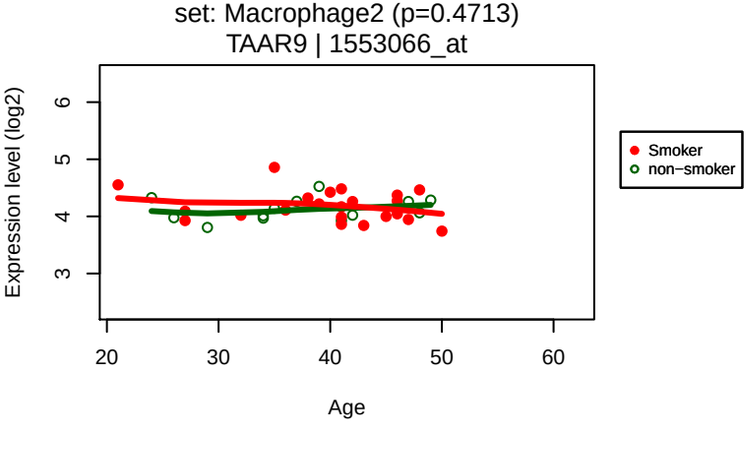

set: SmallAirway2 (p=0.633)
TAAR9 | 1553066_at
Expression level (log2)
Age
Smoker
non−smoker

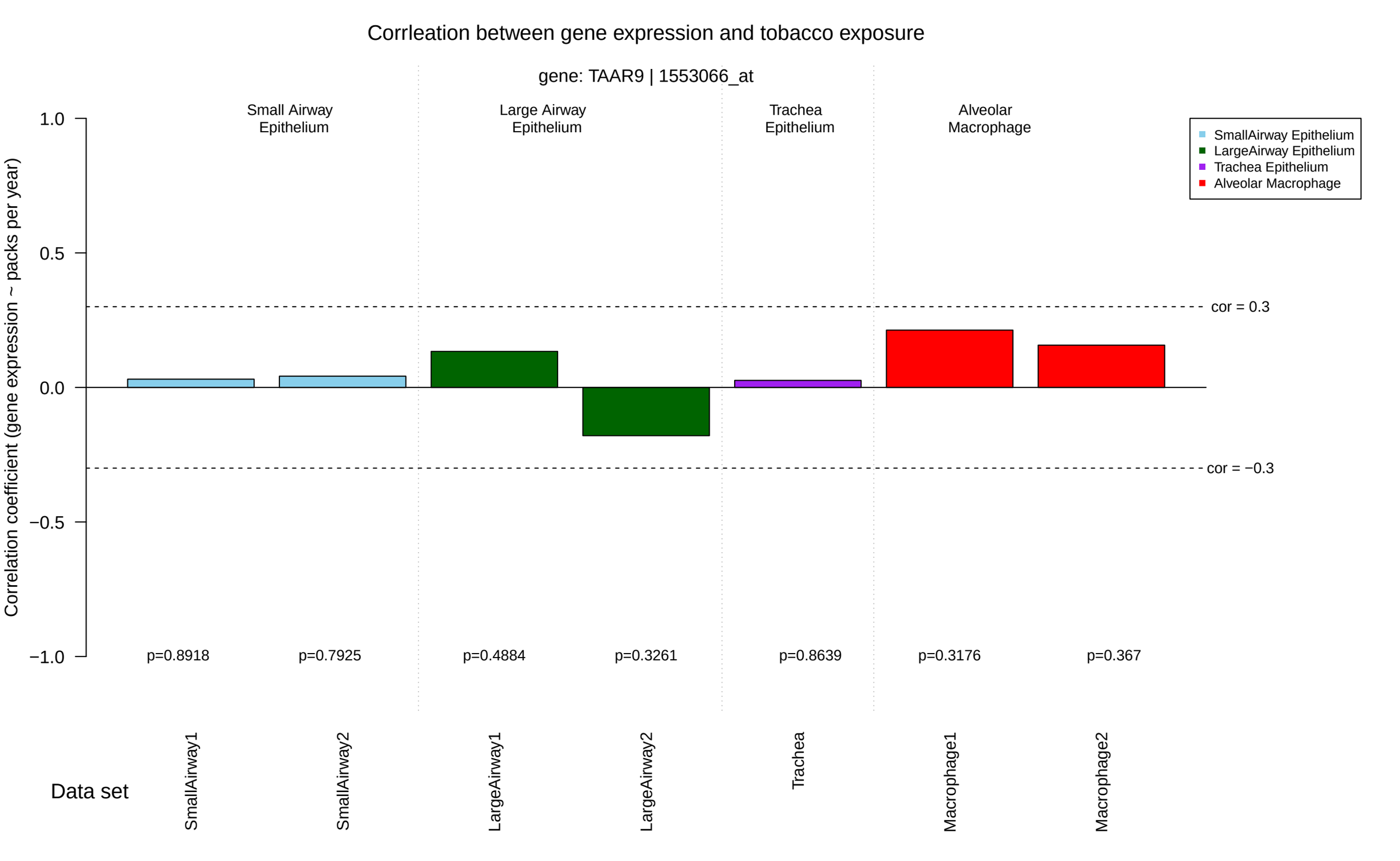

set: LargeAirway1 (p=0.2062)
TAAR9 | 1553066_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1607)
TAAR9 | 1553066_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8039)
TAAR9 | 1553066_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4718)
TAAR9 | 1553066_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4713)
TAAR9 | 1553066_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAAR9 | 1553066_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8918
p=0.7925
p=0.4884
p=0.3261
p=0.8639
p=0.3176
p=0.367
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2